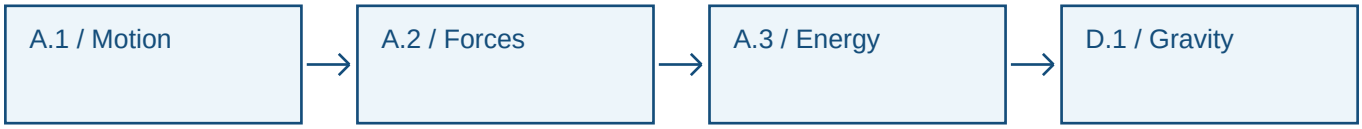
# Physics prerequisite roadmap

A content-based analysis of all five themes, 24 syllabus subtopics and the application skills topic A.0. No example teaching route is used.

## How to read the deductions

Core foundations are inferred from the physics needed to understand a topic. A core permission allows earlier content in a query; it does not mean every individual question requires the whole foundation topic. Conditional links apply only to specific extensions and never expand a topic query automatically.

## Foundation -> dependent topic




Gravity may include earlier circular motion and energy. A forces query does not gain permission for gravity. Gravity also has a direct forces foundation; the diagram shows the transitive chain.

## Dependency layers across the whole syllabus

| Layer | Topics with this longest prerequisite depth |
|---|---|
| 0 | A.0 |
| 1 | A.1 |
| 2 | A.2, A.5 |
| 3 | A.3 |
| 4 | A.4, B.1, B.5, C.1, D.1, D.2 |
| 5 | B.2, B.3, C.2, D.3, D.4 |
| 6 | B.4, C.3, C.5, E.1 |
| 7 | C.4, E.2, E.3 |
| 8 | E.4, E.5 |

Layers are a topological view of the inferred graph, not a prescribed teaching order. Topics in one layer are not prerequisites for one another. The application adds A.0 for measurement and uncertainty skills. It is a user-defined skills category, not an official 2025 IB subtopic code.

# A. Space, time and motion

Begin with measurement and uncertainty skills, describe motion, explain it through forces and conservation, and extend to rotating bodies and reference frames.

## A.0 Measurements and uncertainties [SL / HL skills category]

Measurement methods, SI/base/derived units, dimensional checks, instrument readings and resolution; accuracy, precision, random/systematic errors, significant figures and uncertainty propagation.

**Core: No earlier topic; foundational skills only. Conditional: None assigned.**

Deduction: A cross-cutting skills foundation. Motion and later topics may include measurement/uncertainty skills; selecting A.0 alone does not imply knowledge of later physics topics.

Evidence: TSM printed pp. 2, 7, 9·10. The support material integrates measurement and uncertainty skills throughout physics. A.0 is a user-requested application label, not an official 2025 topic code.

## A.1 Kinematics [SL / HL common]

Position, displacement, velocity and acceleration; motion graphs and constant-acceleration/projectile descriptions.

**Core: A.0. Conditional: None assigned.**

Deduction: Measurement, units, vectors and graphical skills support quantitative descriptions of motion; the application groups these cross-cutting skills under A.0.

Evidence: TSM printed pp. 2, 5, 13. Overview plus standard kinematics concepts inferred from the topic title.

## A.2 Forces and momentum [SL / HL common]

Newton laws, free-body diagrams, equilibrium, momentum and impulse; collisions and centripetal force for circular motion.

**Core: A.1. Conditional: A.3.**

Deduction: Acceleration and velocity underpin dynamics. Energy analysis of elastic/inelastic collisions is a conditional extension, avoiding an A.2/A.3 cycle.

Evidence: TSM printed pp. 3, 6, 17. Explicit conceptual overview; impulse is a standard momentum concept.

## A.3 Work, energy and power [SL / HL common]

Work by a force, kinetic and potential energy, conservation, power and efficiency.

**Core: A.2. Conditional: None assigned.**

Deduction: Work requires force and displacement; mechanical energy builds on motion and dynamics.

Evidence: TSM printed pp. 3, 6, 17. Explicit work-energy overview; power/efficiency deduced from topic scope.

## A.4 Rigid body mechanics [HL only]

Rotational kinematics and dynamics of extended bodies; torque, rotational inertia, rotational energy and angular momentum.

**Core: A.2, A.3. Conditional: None assigned.**

Deduction: Linear equations and Newton laws supply rotational analogies; energy and momentum supply conservation methods.

Evidence: TSM printed pp. 3, 6, 13, 17. Rotational analogies/energy are explicit; torque/inertia are standard deductions.

## A.5 Galilean and special relativity [HL only]

Reference frames, Galilean comparisons and special-relativistic space-time; time dilation and length contraction.

**Core: A.1. Conditional: None assigned.**

Deduction: Comparing motion across frames requires velocity, displacement and time. Nuclear mass-energy use does not require the entire relativity topic.

Evidence: TSM printed pp. 2, 5. Time dilation/length contraction explicit; frame comparisons inferred from title.

# B. The particulate nature of matter

Use microscopic particles and energy transfer to explain thermal behaviour, gases, atmosphere and circuits.

## B.1 Thermal energy transfers [SL / HL common]

Particle models, temperature and internal energy; heat capacity, latent heat, conduction, convection and radiation.

**Core: A.2, A.3. Conditional: None assigned.**

Deduction: Particle motion/forces and kinetic/potential energy explain microscopic storage and macroscopic transfer.

Evidence: TSM printed pp. 3·6, 9·10. Explicit conceptual overview and prior-knowledge example.

## B.2 Greenhouse effect [SL / HL common]

Earth energy balance, absorption and emission of radiation, greenhouse gases and warming.

**Core: B.1. Conditional: C.1, C.4.**

Deduction: Thermal transfer/radiation are core. Molecular resonance is an explanatory extension, not a prerequisite for every energy-balance problem.

Evidence: TSM printed pp. 4·5, 17. Explicit greenhouse/energy-budget description; resonance treated conditionally.

## B.3 Gas laws [SL / HL common]

Ideal-gas model and gas laws; pressure from particle collisions, absolute temperature and molecular kinetic/internal energy.

**Core: B.1, A.2. Conditional: None assigned.**

Deduction: Thermal variables combine with force/momentum and particle motion to connect microscopic and bulk behaviour.

Evidence: TSM printed pp. 4·6, 9·10. Explicit gas model and kinetic-theory discussion.

## B.4 Thermodynamics [HL only]

First and second laws, heat/work/internal energy, gas cycles, heat engines and Carnot efficiency.

**Core: B.3, A.3. Conditional: None assigned.**

Deduction: Gas states and energy accounting underpin cyclic engines. Entropy is a standard second-law concept, not independently detailed here.

Evidence: TSM printed pp. 4, 6, 17. Laws, cycles and efficiency explicit; entropy detail not established by this source.

## B.5 Current and circuits [SL / HL common]

Charge flow and electric current, potential difference as energy per charge, resistance, circuits and electrical power.

**Core: A.3. Conditional: B.1.**

Deduction: Work/energy and rate concepts define voltage and power. Microscopic temperature-resistance models may add thermal content.

Evidence: TSM printed pp. 4·6. Explicit charge flow/resistance/voltage; circuit relations inferred from title.

# C. Wave behaviour

Build from oscillators to propagation, superposition, resonance and motion-dependent observations.

## C.1 Simple harmonic motion [SL + additional HL]

Restoring-force oscillations, displacement and acceleration, kinetic/potential energy exchange and damping.

**Core: A.2, A.3. Conditional: None assigned.**

Deduction: Dynamics defines the restoring acceleration; energy accounts for oscillation and damping.

Evidence: TSM printed pp. 4·6, 17. Explicit oscillation, restoring-force and energy description.

## C.2 Wave model [SL / HL common]

Progressive-wave model; amplitude, wavelength, frequency and speed; mechanical, sound and electromagnetic waves and energy transfer.

**Core: C.1. Conditional: None assigned.**

Deduction: The source explicitly builds the wave model on oscillation. Electromagnetic waves need not import a full fields course.

Evidence: TSM printed pp. 4·5. Oscillation foundation and wave types explicit; basic wave quantities inferred.

## C.3 Wave phenomena [SL + additional HL]

Superposition and interference; wave behaviours including reflection, refraction and diffraction.

**Core: C.2. Conditional: None assigned.**

Deduction: Wavelength, phase and propagation from the wave model underpin overlapping waves and propagation at boundaries.

Evidence: TSM printed pp. 2, 12, 14·15. Superposition/interference explicit; other standard wave phenomena are physics deductions.

## C.4 Standing waves and resonance [SL / HL common]

Standing-wave formation and patterns, harmonics, resonance, driving and damping; vibrating strings and air columns.

**Core: C.1, C.3. Conditional: None assigned.**

Deduction: Superposition creates standing patterns; oscillator response explains resonance.

Evidence: TSM printed pp. 4·6. Standing waves, driving/damping and musical systems explicit; harmonics inferred.

## C.5 Doppler effect [SL + additional HL]

Observed frequency/wavelength changes from source-observer relative motion; sound/light Doppler interpretation and redshift.

**Core: C.2, A.1. Conditional: A.5.**

Deduction: Wave quantities and relative velocity are core. Special-relativistic treatment is conditional and must be separately in scope.

Evidence: TSM printed pp. 2. Topic name supports a physics deduction; detailed Doppler coverage is not provided in this overview.

# D. Fields

Model interactions at a distance and apply force/energy reasoning to masses, charges and induction.

## D.1 Gravitational fields [SL + additional HL]

Universal gravitation, field strength and field maps, orbital/circular motion and Kepler relations; potential, energy and escape.

**Core: A.2, A.3. Conditional: None assigned.**

Deduction: Gravity acts as a force; circular motion uses dynamics; potential and escape use work-energy.

Evidence: TSM printed pp. 4, 6, 11, 17. Explicit overview and progression-of-learning example; some content is additional HL.

## D.2 Electric and magnetic fields [SL + additional HL]

Electric and magnetic field models; Coulomb forces and field lines; electrical potential energy and potential gradients.

**Core: A.2, A.3. Conditional: D.1, B.5.**

Deduction: Force and work-energy supply definitions. Gravity analogy and current-generated field/circuit applications are conditional.

Evidence: TSM printed pp. 4, 6, 17. Explicit force/field/potential descriptions; topic combines SL and additional HL.

## D.3 Motion in electromagnetic fields [SL / HL common]

Charged-particle trajectories in electric/magnetic fields, circular/parabolic paths, combined fields and specific-charge measurements.

**Core: D.2, A.2, A.3. Conditional: B.5.**

Deduction: Field forces and Newtonian trajectories combine; electric acceleration uses energy. Circuit/beam-source apparatus is conditional.

Evidence: TSM printed pp. 5·6, 11·12, 17. Explicit conceptual overview and assessment-learning example.

## D.4 Induction [HL only]

Changing magnetic flux and induced emf/current; Faraday/Lenz reasoning, electricity generation and energy conservation.

**Core: D.2, B.5, A.3. Conditional: None assigned.**

Deduction: Magnetic field/flux, circuit quantities and work-energy are foundations; Lenz direction enforces conservation.

Evidence: TSM printed pp. 4, 17. Lenz/generation explicit; flux and Faraday relation are standard induction deductions.

# E. Nuclear and quantum physics

Connect atomic/nuclear structure, quantization, radiation and nuclear energy with microscopic evidence and stellar observations.

## E.1 Structure of the atom [SL + additional HL]

Rutherford nuclear atom, electrons/protons/neutrons; Bohr quantized levels and absorption/emission spectra.

**Core: D.2, C.2, A.3. Conditional: None assigned.**

Deduction: Scattering/closest approach use field forces and energy; spectra connect photon energy with wave frequency.

Evidence: TSM printed pp. 4·5, 12, 17. Explicit overview and atom-model example; mixed SL/additional HL depth.

## E.2 Quantum physics [HL only]

Photoelectric energy balance, wave-particle duality, de Broglie matter waves and Compton scattering.

**Core: E.1, C.2, A.2. Conditional: B.5, C.3.**

Deduction: Atomic/photon energy and wave quantities combine with momentum. Stopping-potential circuits and diffraction evidence are conditional.

Evidence: TSM printed pp. 4, 6, 17. Photoelectric/de Broglie/Compton explicit; experimental prerequisites deduced.

## E.3 Radioactive decay [SL + additional HL]

Nuclear composition/stability, short-range nuclear force, radioactive transformations, decay/half-life and binding-energy mass changes.

**Core: E.1, A.3. Conditional: None assigned.**

Deduction: Atomic nuclear structure and conservation/energy underpin nuclear change; the whole A.5 course is not necessary for a supplied mass-energy relation.

Evidence: TSM printed pp. 4, 7, 17. Nuclear forces/binding energy explicit; decay/half-life are standard deductions from title.

## E.4 Fission [SL / HL common]

Fission of heavy nuclei, chain reactions, nuclear energy release and nuclear-power generation.

**Core: E.3. Conditional: B.1, B.4.**

Deduction: Binding-energy and nuclear-reaction ideas are core. Thermal plant operation and engine efficiency are conditional.

Evidence: TSM printed pp. 4, 17. Fission/power and mass-energy explicit; chain reactions inferred from content scope.

## E.5 Fusion and stars [SL / HL common]

Fusion as a stellar energy source, stellar observations/spectra and thermal radiation; temperature, luminosity and stellar models.

**Core: E.3, B.1. Conditional: D.1, B.3, C.5, E.4.**

Deduction: Nuclear energetics and thermal radiation are core. Gravitational equilibrium, gas models, Doppler and comparison with fission are conditional.

Evidence: TSM printed pp. 4, 10, 17. Fusion/stars, spectra, brightness, temperature and radius explicit. This is not an exhaustive astrophysics syllabus.

# Rules for topic-safe selection

Use the content prerequisites together with a separately approved subquestion dependency map.

### 1. Expand only core foundations

Allowed scope = selected targets plus their transitive core foundations. Conditional links are excluded unless selected explicitly or already reached through a core path. Never infer reverse links or all same-theme topics.

### 2. Require a target task

At least one retained task must be tagged with a selected target. A mechanics-only question is not a gravity result, even though mechanics is in the gravity knowledge scope.

### 3. Remove out-of-scope extensions

For A.1 motion, remove an optional A.3 energy extension. For D.1 gravity, allow A.2 circular motion; for A.2 forces, do not permit D.1. An inseparable multi-topic task needs all its topics in the allowed scope.

### 4. Preserve the dependency direction

If X requires A, retaining X always retains A and the shared stem. Retaining A does not force X. When an out-of-scope prerequisite is removed, remove its dependants too. Reject the parent if no target task survives or the shared stem is out of scope.

### 5. Keep approval and level checks

Only current, reviewed part crops, topic tags and acyclic dependency maps authorize trimming. Missing or stale maps permit a whole question only if all parent tags fit. HL/SL suitability is a separate check; mixed-depth topic codes do not certify SL content.

### 6. Apply Full and Short consistently

Unfiltered Full preserves every crop. Topic-filtered Full keeps every in-scope task. Short may additionally omit optional tasks, preferring independent tasks, while preserving a target task and all required context.

## Source and limits

IB Physics teacher support material, first assessment 2025; published September 2023, updated July/November 2024. Analysis uses printed pp. 2-17 (PDF pp. 7-22); the sample teaching route is excluded. The uploaded PDF has 116 pages. Full source identity and per-topic rationales are in curriculum_prerequisites.json.

This is a content-based project synthesis, not an official IB prerequisite specification or an exhaustive syllabus checklist. The support material is brief on some topics; the cards label physics deductions where details are inferred. Core versus conditional is an application judgement. A.0 is a user-defined cross-cutting skills category, not an official IB 2025 subtopic code.

Small-model dependency suggestions require human approval. Current real inference has not demonstrated reliable dependency classification; conservative validation rejects invalid graphs.